



Imperial College Healthcare  
NHS Trust

**iCARE**

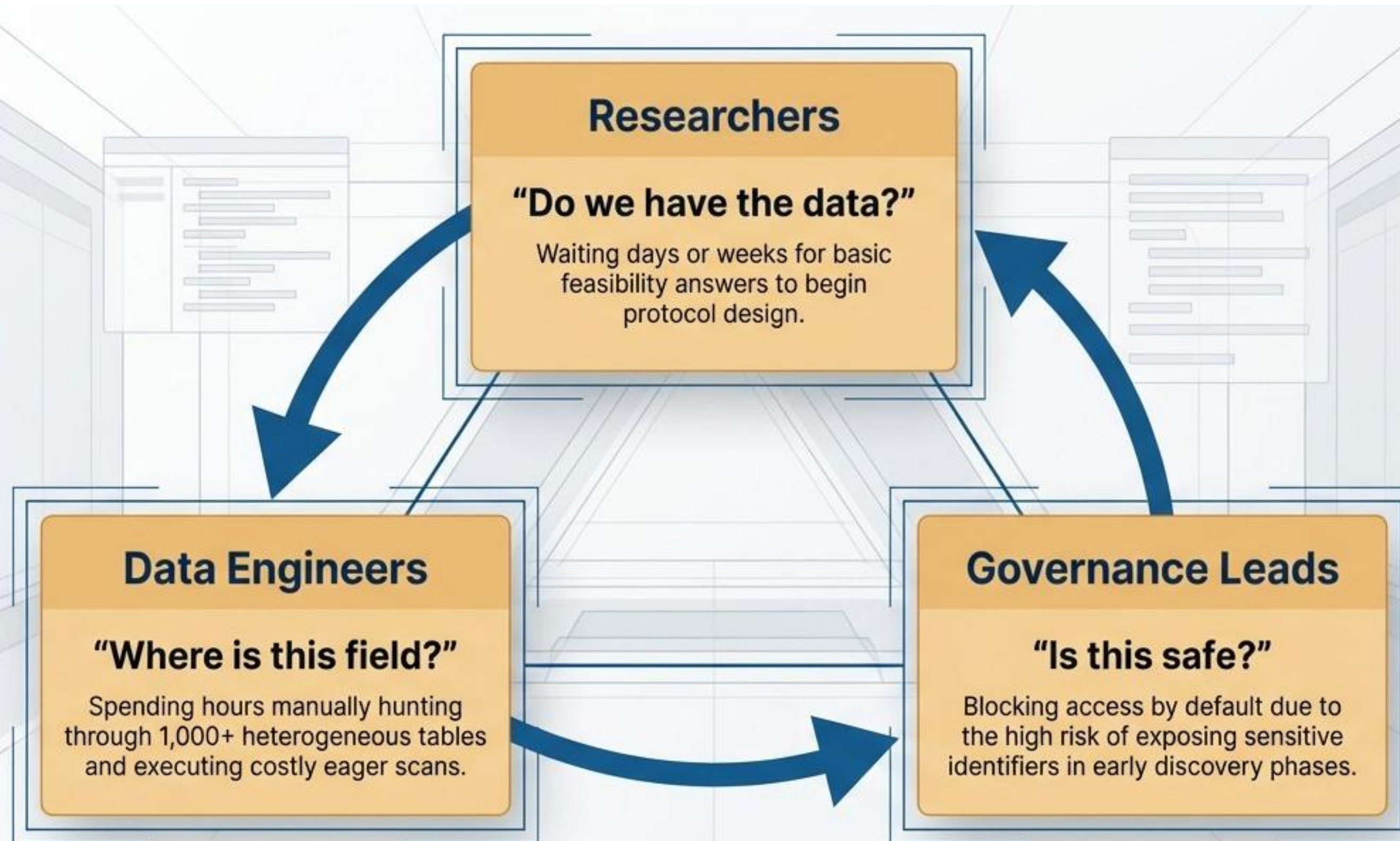
**Illuminating the Dark Data**

# Data Champs

Fast, role-aware data feasibility for  
clinical research teams.

```
> datachamps analyze --target="clinical_estate" [READY]
```

# Clinical analytic cycle



# Use cases



## Protocol Feasibility

Rapidly determine whether required data elements exist across heterogeneous clinical systems.



## Cohort Discovery

Identify viable patient populations without waiting weeks for manual data pulls.



## Data Quality Triage

Quickly assess whether fields are complete, consistent, and usable before committing analyst time.



## Study Startup Acceleration

Give clinical teams immediate visibility into data readiness to reduce back-and-forth with data engineering.



## Governance-Safe Exploration

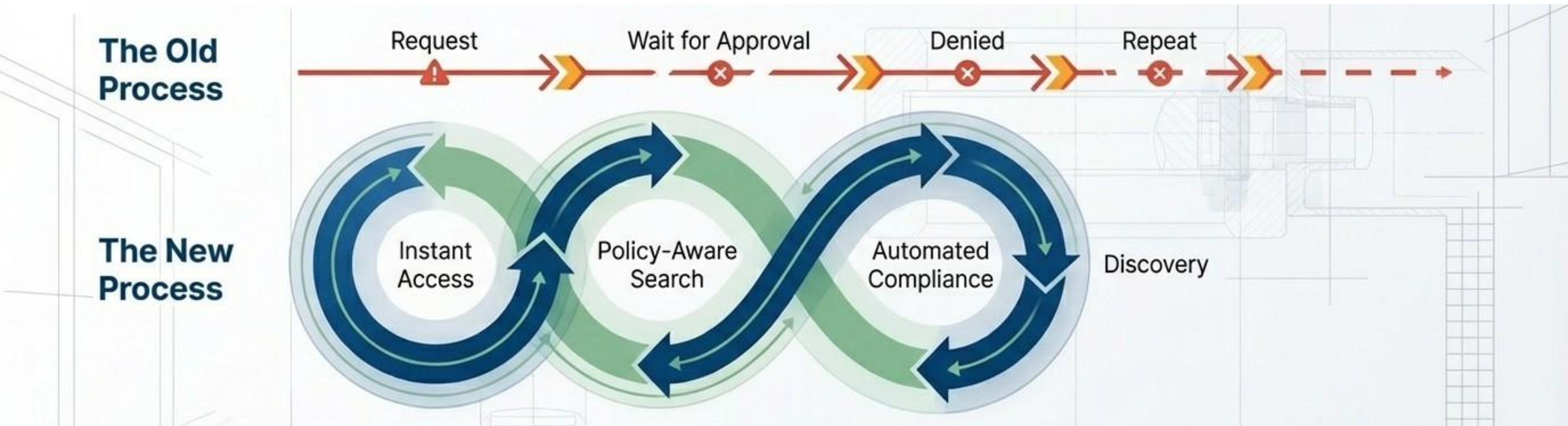
Enable role-aware, metadata-only discovery without exposing row-level data.



# Impact - Transferring feasibility from Weeks to Minutes

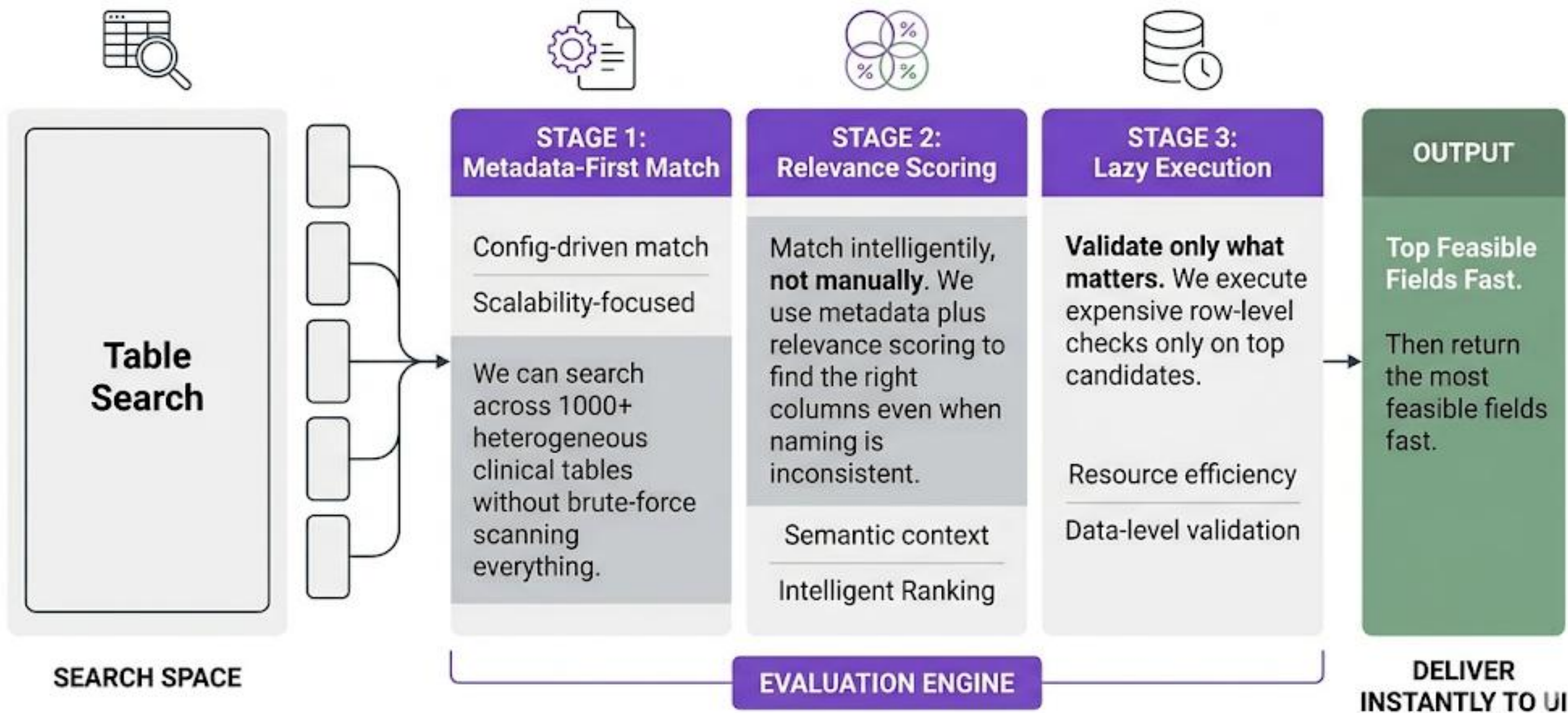
|            | Manual Feasibility               | Data Champs                                      |
|------------|----------------------------------|--------------------------------------------------|
| Time       | Weeks of back-and-forth tickets. | Minutes for a first-pass shortlist.              |
| Compute    | Brute-force eager full scans.    | Lazy execution triggering only on exact matches. |
| Governance | Post-facto compliance reviews.   | Role-Based Access Control (RBAC) by design.      |
| Interface  | Manual SQL querying.             | Guided Streamlit UI & CLI layer.                 |

# Governance as an Accelerator, not a Brake





# Technical approach



# Technical approach – Streamlit APP

100 Tables Scanned

Inter

50 Concepts Searched

Inter

2

GREEN: Ready  
for extraction.

Inter

1

AMBER: Feasible  
with caveats.

Inter

47

RED: Insufficient data.

Inter

## Diagnostic Insight

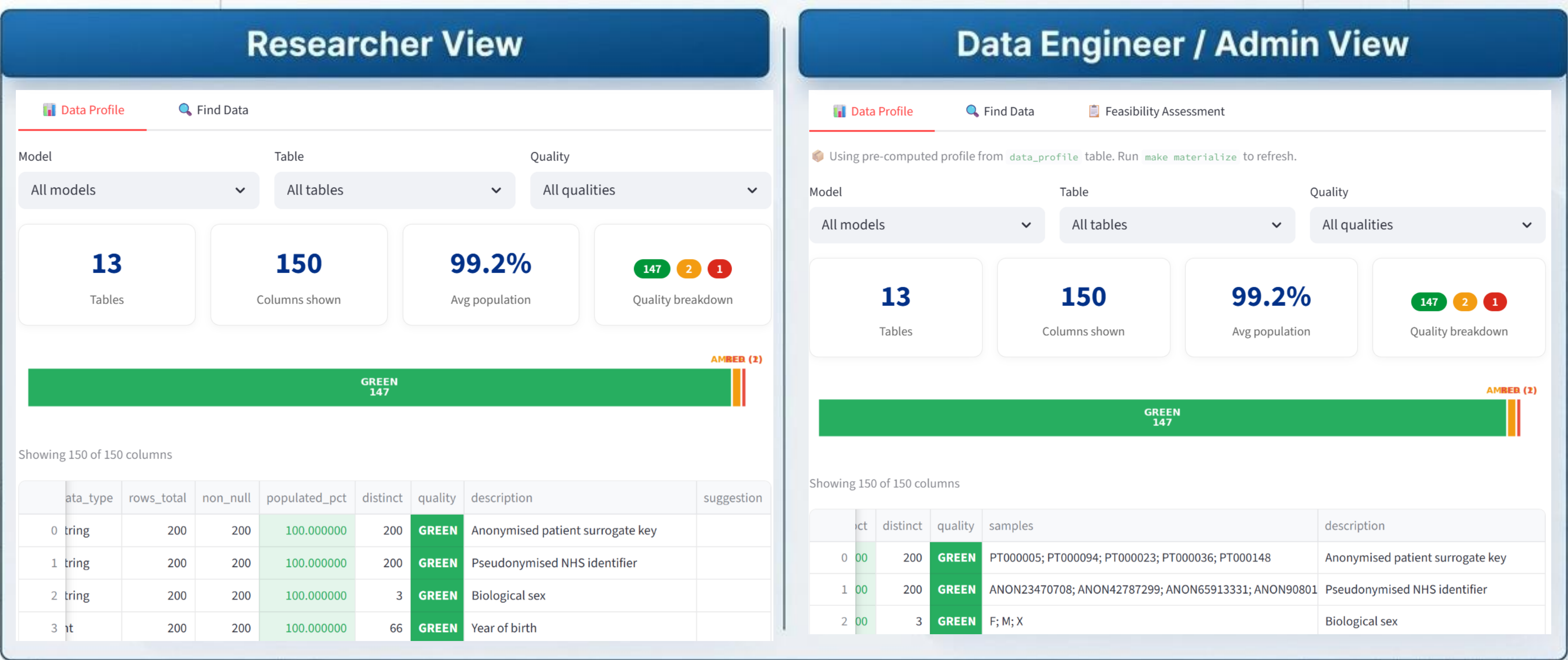
A high RED rate is a success metric. It provides an instant mapping of data coverage gaps requiring remediation, saving weeks of wasted engineering time hunting for data that doesn't exist.

## RBAC Policy Output

|               |     |             |     |
|---------------|-----|-------------|-----|
| 👤 FULL_ACCESS | (0) | 👤 ANON_ONLY | (2) |
| 🚫 NO_ACCESS   | (0) | 🚫 BLOCKED   | (0) |



# The Streamlit APP - Tailored visibility by Role





# The Streamlit APP - Tailored visibility by Role

 Settings

Your role

oncologist

 **Oncologist**

Full access to treatment, MDT, radiotherapy & systemic therapy tables.

Governance rules

- Restricted columns: `subject`, `encntr_id`, `episode_id`
- Allowed database: `icht_prod`
- Sample values: only for `data_engineer` / `admin`

Accessible tables

Deploy

 **Data Champs — Data Profiling Tool**

Imperial College Healthcare NHS Trust · RBAC-governed data profiling & discovery

 Data Profile  Find Data

Describe what data you need in **plain English**. We search across **all** tables — including ones you may need to request access to.

Patient smoking history and treatment outcomes

☒ Smart ☐ Keyword

 1 matches  0 accessible  1 require permission

▼ raw\_cancer\_patients . smoking\_status — relevance: 0.454

 Request access  Model: cancer\_model · Type: string

 **Description:** Smoking history (never, former, current)

| Rows | Non-null | Populated |
|------|----------|-----------|
| 200  | 200      | 100.0%    |

# Production feasibility

